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In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

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should indicates a recommendation to do something

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need not indicates permission not to do something

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can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the stage 3 protocol and data model for the Nadm Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the ADM.

The 5G System stage 2 architecture and procedures are specified in TS 23.501 [2], TS 23.502 [3] and TS 23.369[14].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in TS 29.500 [4] and TS 29.501 [5].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".

[4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".

[5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".

[6] OpenAPI: "OpenAPI Specification Version 3.0.0", https://spec.openapis.org/oas/v3.0.0.

[7] 3GPP TR 21.900: "Technical Specification Group working methods".

[8] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[9] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".

[10] 3GPP TS 29.510: "5G System; Network Function Repository Services; Stage 3".

[11] IETF RFC 9113: "HTTP/2".

[12] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[13] IETF RFC 9457: "Problem Details for HTTP APIs".

[14] 3GPP TS 23.369: "Architecture support for Ambient power-enabled Internet of Things; Stage 2".

[15] IETF RFC 9110: "HTTP Semantics".

[16] IETF RFC 9111: "HTTP Caching".

[17] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces Stage 3".

[18] 3GPP TS 33.369: " Security aspects of Ambient Internet of Things (AIoT) services for isolated private networks".

[19] 3GPP TS 29.503: "5G System; Unified Data Management Services; Stage 3".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and TS 23.369 [14] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and TS 23.369 [14] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

4 Overview

Within the 5GC, the ADM offers services to the AIOTF, NEF via the Nadm service based interface (see TS 23.369 [14]).

Figure 4-1 provides the reference model (in service based interface representation and in reference point representation), with focus on the ADM:

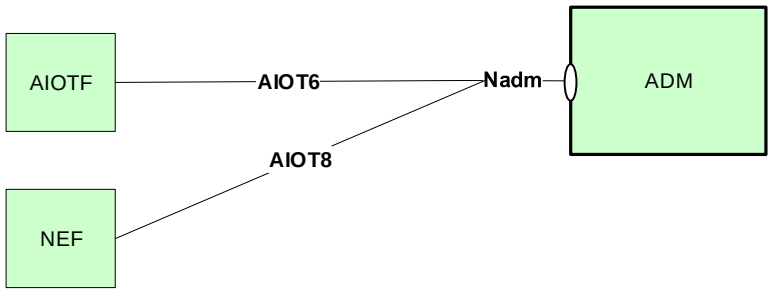


Figure 4-1: Reference model – ADM

The functionalities supported by the ADM are listed in clause 4.5.9 of TS 23.369 [14].

5 Services offered by the ADM

5.1 Introduction

The ADM offers the following services via the Nadm interface:

- Nadm_DM Service
- Nadm_Sec

Table 5.1-1 summarizes the corresponding APIs defined for this specification.

Table 5.1-1: API Descriptions

Service Name	Clause	Description	OpenAPI Specification File	apiName	Annex
Nadm_DM	6.1	ADM Data Management	TS29369_Nadm_DM.yaml	nadm-dm	A.2
Nadm_Sec	6.2	ADM Security Service	TS29369_Nadm_Sec.yaml	nadm-sec	A.3

5.2 Nadm_DM Service

5.2.1 Service Description

The Nadm_DM service enables an NF to request AIoT device profile data or the AF authorization data or update the AIoT device profile data in the ADM.

5.2.2 Service Operations

5.2.2.1 Introduction

The service operations defined for the Nadm_DM service are as follows:

- Query: It enables a consumer NF to request AIoT device profile data or the AF authorization data from the ADM.
- Update: It enables a consumer NF to update the AIoT device profile data in the ADM.
- Subscribe: It enables a consumer NF to create or modify a subscription for AF Authorization Data changes
- Unsubscribe: It enables a consumer NF to remove a subscription for AF Authorization Data changes
- Notify: It enables a consumer NF to receive a notification for AF Authorization Data changes.

5.2.2.2 Query

5.2.2.2.1 General

The following procedures using the Query service operation are supported:

- AIoT Device Profile Data Retrieval
- AF Authorization Data Retrieval
- Bundled AIoT Device Profile Data Retrieval

5.2.2.2.2 AIoT Device Profile Data Retrieval

Figure 5.2.2.2.2-1 shows a scenario where the NF service consumer (e.g. AIOTF or NEF) sends a request to the ADM to receive the AIoT Device Profile Data of a given AIoT device (see TS 23.369 [14]).

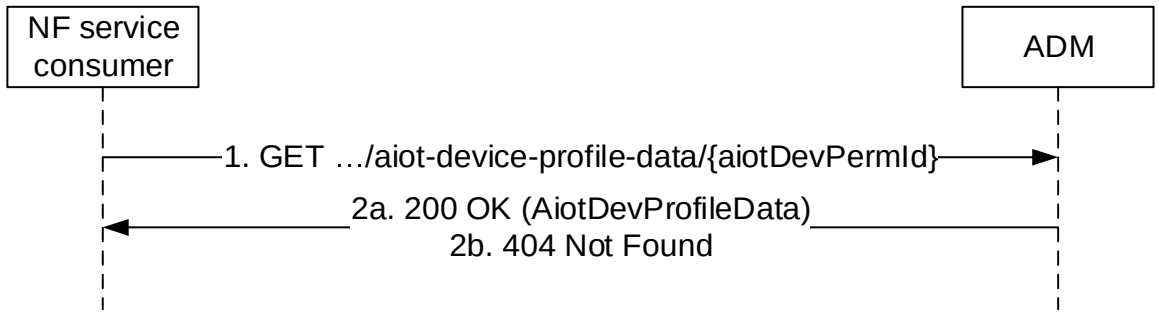


Figure 5.2.2.2.2-1: Requesting AIoT Device Profile Data

1. The NF service consumer (e.g. AIOTF or NEF) sends a GET request to the resource …/aiot-device-profile-data/{aiotDevPermId}, to get the AIoT Device Profile Data of a given AIoT device.

2a. On success, the ADM responds with "200 OK" with the AIoT Device Profile Data.

2b. If there is no valid AIoT Device Profile Data for the AIoT device permanent identifier, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the GET response body.

5.2.2.2.3 AF Authorization Data Retrieval

Figure 5.2.2.2.3-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to receive the AF Authorization Data (see TS 23.369 [14]).

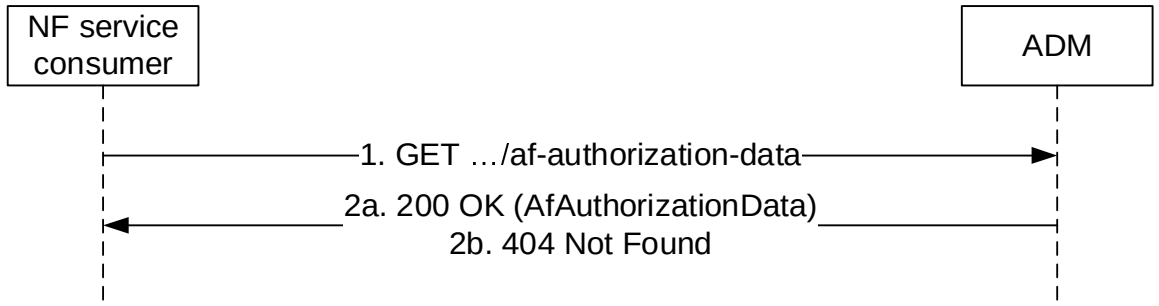


Figure 5.2.2.2.3-1: Requesting AF Authorization Data

1. The NF service consumer (e.g. AIOTF) sends a GET request to the resource of the AF authorization data (…/af-authorization-data), to get the Authorization Data of the AFs.

   If the retrieval is targeting a specific AF authorization data, the request shall contain the target AF ID. Otherwise, the retrieval is to fetch all the stored AF authorization data.

2a. On success, the ADM responds with "200 OK" with the Authorization Data of the target AF(s).

2b. If there is no valid AF Authorization Data available, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the GET response body.

5.2.2.2.4 Bundled AIoT Device Profile Data Retrieval

Figure 5.2.2.2.4-1 shows a scenario where the NF service consumer (e.g. AIOTF or NEF) sends a request to the ADM to receive the AIoT Device Profile Data for bundled AIoT Devices (see TS 23.369 [14]).

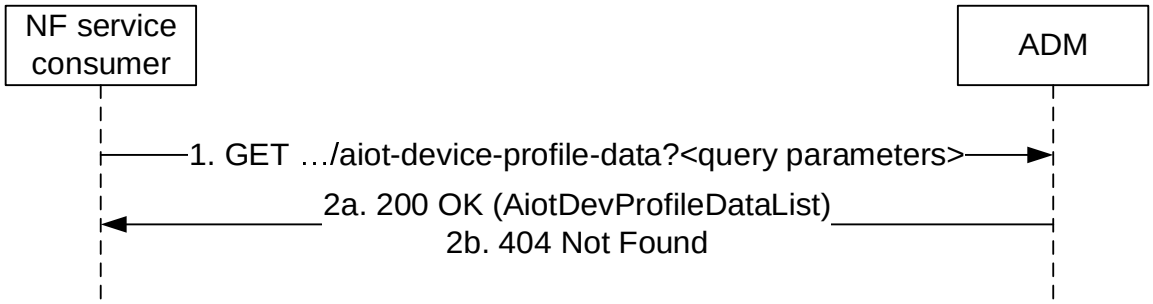


Figure 5.2.2.2.4-1: Requesting AIoT Device Profile Data for bundled AIoT Devices

1. The NF service consumer (e.g. AIOTF or NEF) sends a GET request to the resource URI "aiotDevPermId" collection resource. The optional input filter criteria (a list of aiotDevPermId) for the retrieval request may be included in the query parameters. All the aiotDevPermId included in the list are aggregated by the NF service consumer (e.g. AIOTF) based on the the list of device IDs served by ADM, which can be retrieved from NRF and included in the deviceIdList within AdmInfo.

   The NF service consumer (e.g. AIOTF) should not include the aiotDevPermId that are not served by the ADM and after aggregation, the NF service consumer (e.g. AIOTF) shall bundle appropriate number of AIoT devices based on deployment to avoid bundling huge number of requested devices.

2a. On success, the ADM responds with "200 OK". The response body shall contain the URI (conforming to the resource URI structure as described in clause 5.2.2.2.2) of each aiotDevPermId in the ADM that satisfy the retrieval filter criteria (a list of aiotDevPermId).

2b. If there is no valid AIoT Device Profile Data for all the AIoT device permanent identifiers listed in the input filter criteria, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the GET response body.

5.2.2.3 Update

5.2.2.3.1 General

The following procedures using the Update service operation are supported:

- AIoT Device Profile Data Update
- Bundled AIoT Device Profile Data Update

5.2.2.3.2 AIoT Device Profile Data Update

Figure 5.2.2.3.2-1 shows a scenario where the NF service consumer (e.g., AIOTF) sends a request to the ADM to modify the AIoT Device Profile Data of a given AIoT device (see TS 23.369 [14]). The request contains the AIoT device permanent identifier and the modification instructions.

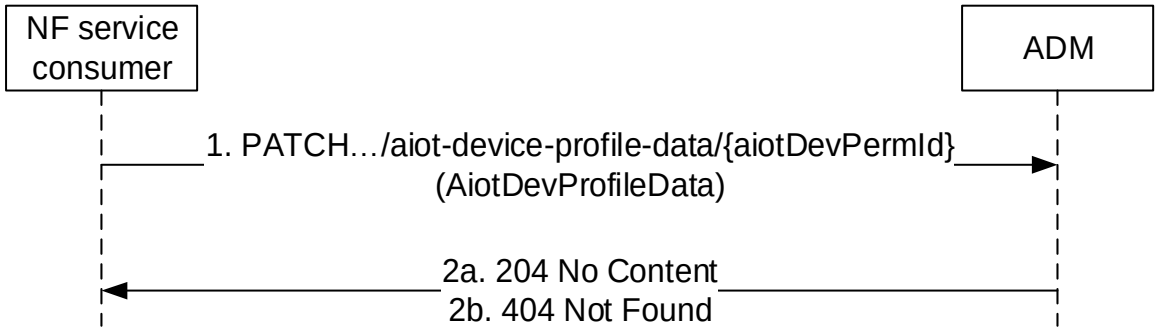


Figure 5.2.2.3.2-1: Updating AIoT Device Profile Data

1. The NF service consumer (e.g. AIOTF) sends a PATCH request to the resource …/aiot-device-profile-data/{aiotDevPermId}, to update the AIoT Device Profile Data of a given AIoT device.

2a. On success, the ADM responds with "204 No Content".

2b. If there is no valid AIoT Device Profile Data for the AIoT device permanent identifier, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the PATCH response body.

5.2.2.3.3 Bundled AIoT Device Profile Data Update

Figure 5.2.2.3.3-1 shows a scenario where the NF service consumer (e.g., AIOTF) sends a request to the ADM to modify the AIoT Device Profile Data for a bundled AIoT Devices (see TS 23.369 [14]). The request contains the the modification instructions.

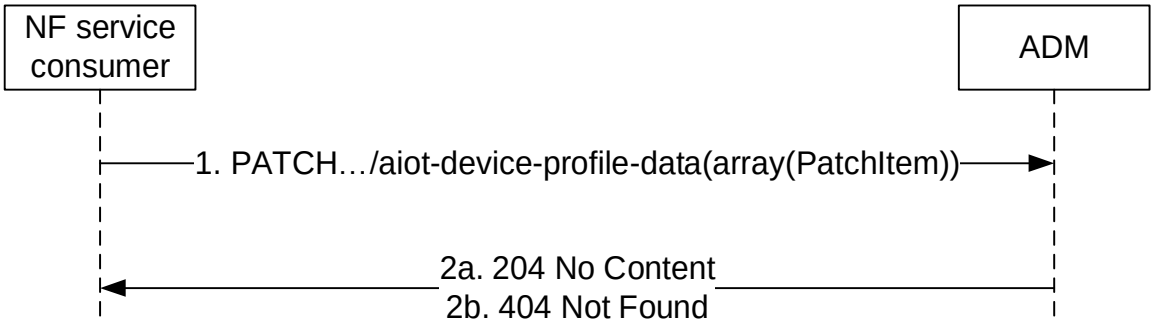


Figure 5.2.2.3.3-1: Updating AIoT Device Profile Data for bundled AIoT Devices

1. The NF service consumer (e.g. AIOTF) sends a PATCH request to the resource URI "aiotDevPermId" collection resource. The content of the PATCH request shall contain the list of operations (add/delete/replace) to be applied to the AIoT Device Profile Data for bundled AIoT Devices, these operations may be directed to bundled AIoT Devices, where each of them directed to individual AIoT Device Profile Data parameters for a given AIoT device permanent identifier or to all parameters offered by a given AIoT Device permanent identifier.

2a. On success, the ADM responds with "204 No Content".

2b. If there is no valid AIoT Device Profile Data for the AIoT device permanent identifier, HTTP status code "404 Not Found" shall be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the PATCH response body.

5.2.2.4 Subscribe

5.2.2.4.1 General

The following procedures using the Subscribe service operation are supported:

- Create a subscription of AF Authorization Data change
- Modify a subscription of AF Authorization Data change

5.2.2.4.2 Create a subscription of AF Authorization Data change

Figure 5.2.2.4.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to create a subscription for AF Authorization Data change (see TS 23.369 [14]).

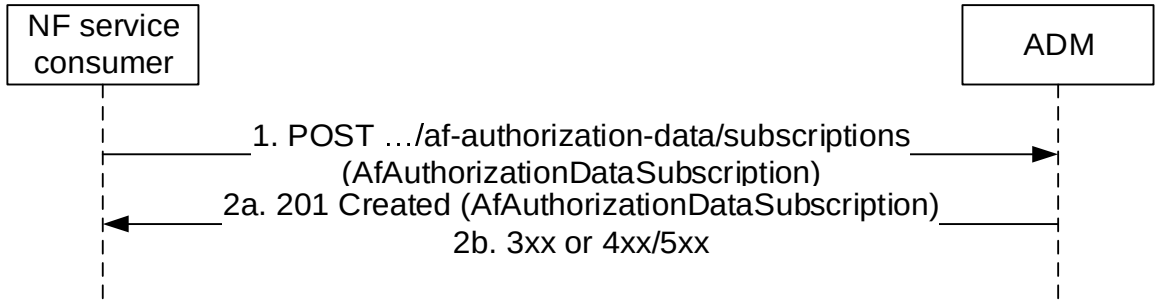


Figure 5.2.2.4.2-1: Create a subscription of AF Authorization Data change

1. The NF service consumer (e.g. AIOTF) sends a POST request to the resource of the AF authorization data subscriptions (…/af-authorization-data/subscriptions), to subscribe to the changes of Authorization Data of the AFs. The request may contain the target AF ID if the changes of Authorization data for a specific AF is to be monitored.

2a. On success, the ADM responds with "201 Created" with created subscription to the Authorization Data of the target AF(s). The "Location" header shall be included in the response and shall include the resource URI of the newly created subscription.

2b. On redirection or failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.2.2.4.3 Modify a subscription of AF Authorization Data change

Figure 5.2.2.4.3-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to modify a subscription for AF Authorization Data change (see TS 23.369 [14]).

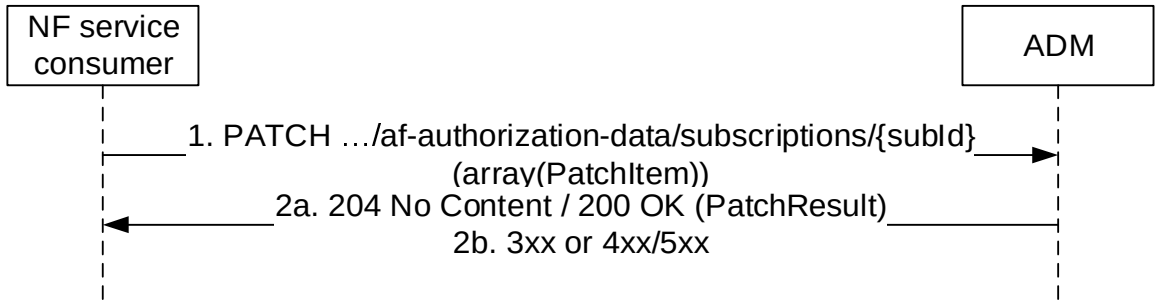


Figure 5.2.2.4.3-1: Modify a subscription of AF Authorization Data change

1. The NF service consumer (e.g. AIOTF) sends a PATCH request to the resource of the individual AF authorization data subscription (…/af-authorization-data/subscriptions/{subId}), to modify a subscription for Authorization Data change of the AFs.

2a. On success, the ADM responds with "204 No Content" shall be returned. On partial success, "200 OK" shall be returned with the patch result.

2b. On redirection or failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.2.2.5 Unsubscribe

5.2.2.5.1 General

The following procedures using the Unsubscribe service operation are supported:

- Delete a subscription of AF Authorization Data change

5.2.2.5.2 Delete a subscription of AF Authorization Data change

Figure 5.2.2.5.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to delete a subscription for AF Authorization Data change (see TS 23.369 [14]).

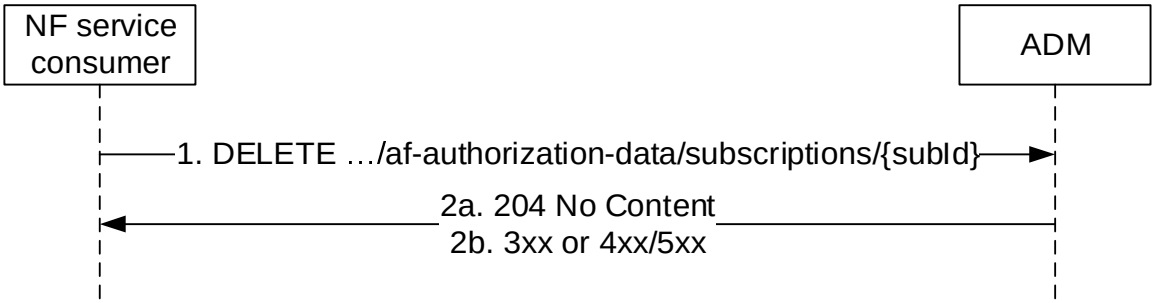


Figure 5.2.2.5.2-1: Delete a subscription of AF Authorization Data change

1. The NF service consumer (e.g. AIOTF) sends a DELETE request to the resource of the individual AF authorization data subscription (…/af-authorization-data/subscriptions/{subId}), to delete a subscription for Authorization Data change of the AFs.

2a. On success, the ADM responds with "204 No Content ".

2b. On redirection or failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.2.2.6 Notify

5.2.2.6.1 General

The following procedures using the Notify service operation are supported:

- Notify changes of AF Authorization Data

5.2.2.6.2 Notify changes of AF Authorization Data

Figure 5.2.2.6.2-1 shows a scenario where the ADM sends a request to the NF service consumer (e.g. the AIOTF) to notify the changes of AF Authorization Data (see TS 23.369 [14]).

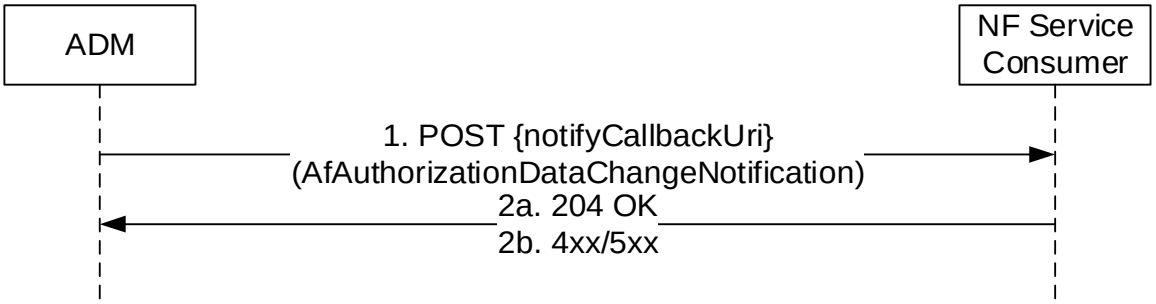


Figure 5.2.2.6.2-1: Notify changes of AF Authorization Data

1. The ADM sends a POST request towards the callback reference provided by the NF service consumer ({notifyCallbackUri}), to notify the NF service consumer for the changes of Authorization Data change of the AFs.

2a. On success, the ADM responds with "204 No content".

2b. On redirection or failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3 Nadm_Sec Service

5.3.1 Service Description

See TS 33.369 [18].

5.3.2 Service Operations

5.3.2.1 Introduction

For the Nadm_Sec service the following service operations are defined:

- RAND_Get
- Authentication_Get
- SessionKey_Get
- TID_Get
- TID_Update

The Nadm_Sec Service is used by the AIOTF to request the ADM to provide a random number (RAND), authentication data for a single AIoT device or a group of AIoT devices, the session key for an AIoT device, the T-ID for an AIoT device, and to update the T-ID for an AIoT device.

5.3.2.2 RAND_Get

5.3.2.2.1 General

The following procedures using the RAND_Get service operation are supported:

- RAND Retrieval

5.3.2.2.2 RAND Retrieval

Figure 5.3.2.2.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to receive a random number (see TS 33.369 [18]).

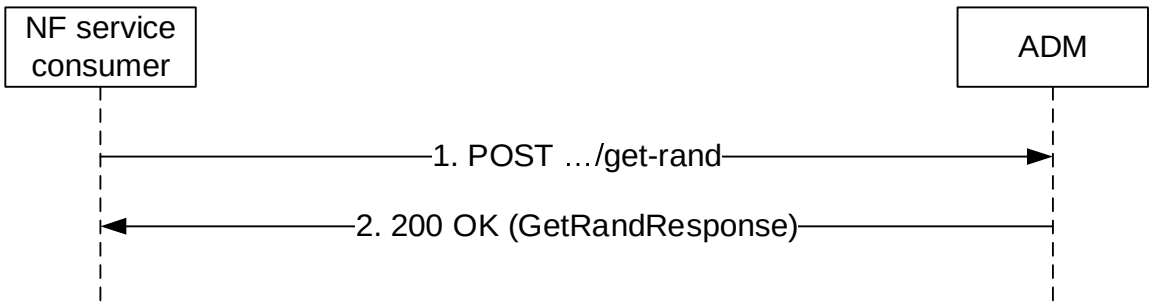


Figure 5.3.2.2.2-1: RAND Retrieval

1. The NF service consumer sends a POST request (custom method: get-rand).
2. The ADM responds with "200 OK" with the message body containing the generated random number.

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3.2.3 Authentication_Get

5.3.2.3.1 General

The following procedures using the Authentication_Get service operation are supported:

- Authentication Data Retrieval

5.3.2.3.2 Authentication Data Retrieval

Figure 5.3.2.3.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to receive the authentication data for a single AIoT device or a group of AIoT devices (see TS 33.369 [18]).

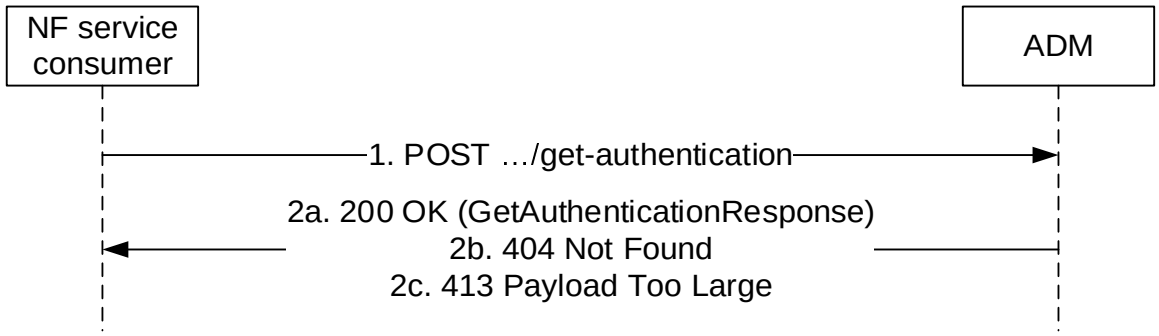


Figure 5.3.2.3.2-1: Authentication Data Retrieval

1. The NF service consumer sends a POST request (custom method: get-authentication) to the ADM, including authentication input parameters for a single AIoT Device, identified by an AIoT device permanent identifier, or a group of AIoT Devices, identified by Filter Information.

2a. The ADM responds with "200 OK" with the message body containing the derived authentication data.

2b. If the AIoT device permanent identifier or Filtering Information is not recognized, HTTP status code "404 Not Found" may be returned including additional error information in the response body (in the "ProblemDetails" element).

2c. If the group of devices derived from the Filtering Information is too large to compute the expected authentication data ($XRES_{AIOT}$ values) in a reasonable amount of time, HTTP status code "413 Payload Too Large" may be returned. The ADM shall include the application error "FILTER_INFORMATION_TOO_BROAD" in the ProblemDetails element to indicate that the filtering criteria resulted in an unmanageable device set.

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3.2.4 SessionKey_Get

5.3.2.4.1 General

The following procedures using the SessionKey_Get service operation are supported:

- Session Key Retrieval

5.3.2.4.2 Session Key Retrieval

Figure 5.3.2.4.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to receive the Session Key for an AIoT device (see TS 33.369 [18]).

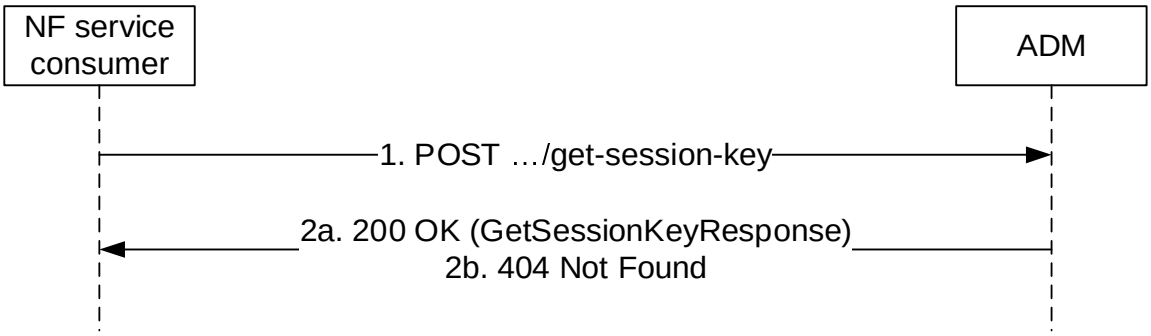


Figure 5.3.2.4.2-1: Session Key Retrieval

1. The NF service consumer sends a POST request (custom method: get-session-key) to the ADM, including input parameters for an AIoT Device, identified by an AIoT device permanent identifier.

2a. The ADM responds with "200 OK", with the message body containing the derived Session Key.

2b. If the AIoT device permanent identifier is not recognized, HTTP status code "404 Not Found" may be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3.2.5 TID_Get

5.3.2.5.1 General

The following procedures using the TID_Get service operation are supported:

- TID Retrieval

5.3.2.5.2 TID Retrieval

Figure 5.3.2.5.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to receive the TID (see TS 33.369 [18]).

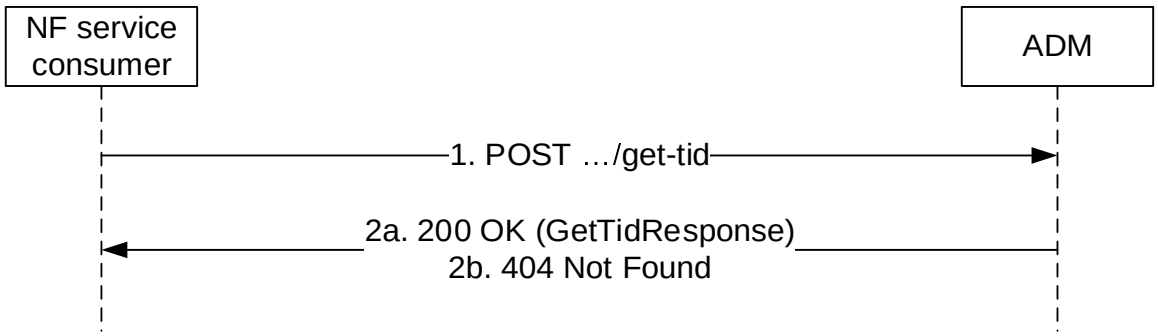


Figure 5.3.2.5.2-1: TID Retrieval

1. The NF service consumer sends a POST request (custom method: get-tid) to the ADM, including input parameters such as the AIoT device permanent identifier and, optionally, a resynchronization indicator and $RAND_{AIOT_n}$.

2a. The ADM responds with "200 OK", with the message body containing the allocated Temporary Identifier (T-ID) and the associated T-ID handling information.

2b. If the AIoT device permanent identifier is not recognized, HTTP status code "404 Not Found" may be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

5.3.2.6 TID_Update

5.3.2.6.1 General

The following procedures using the TID_Update service operation are supported:

- TID Update

5.3.2.6.2 TID Update

Figure 5.3.2.6.2-1 shows a scenario where the NF service consumer (e.g. AIOTF) sends a request to the ADM to generate a new value for the stored type T-ID (see TS 33.369 [18]).

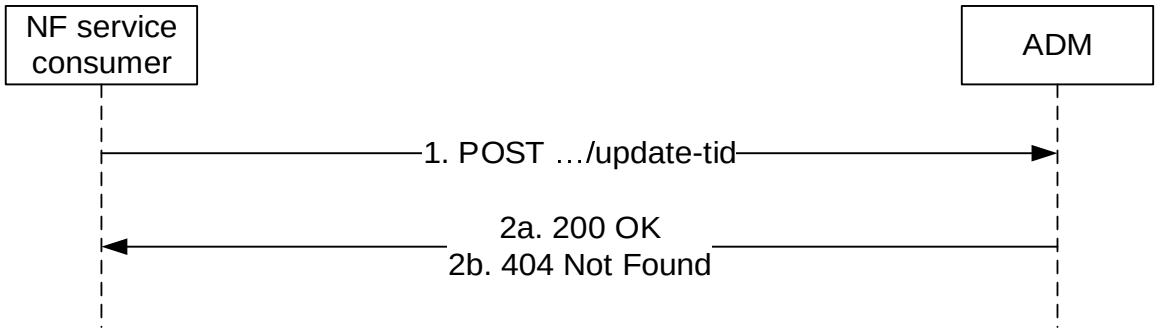


Figure 5.3.2.6.2-1: TID Update

1. The NF service consumer sends a POST request (custom method: update-tid) to the ADM, including input parameters such as the AIoT device permanent identifier and $RAND_{AIOT_n}$.

2a. The ADM responds with "200 OK", with the message body containing the updated Temporary ID (T-ID).

2b. If the AIoT device permanent identifier is not recognized, HTTP status code "404 Not Found" may be returned including additional error information in the response body (in the "ProblemDetails" element).

On failure, the appropriate HTTP status code indicating the error shall be returned and appropriate additional error information should be returned in the POST response body.

6 API Definitions

6.1 Nadm_DM Service API

6.1.1 Introduction

The Nadm_DM shall use the Nadm_DM API.

The API URI of the Nadm_DM API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nadm-dm".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

6.1.2 Usage of HTTP

6.1.2.1 General

HTTP/2, IETF RFC 9113 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nadm_DM API is contained in Annex A.

6.1.2.2 HTTP standard headers

6.1.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

6.1.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 9457 [13].

6.1.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

6.1.3 Resources

6.1.3.1 Overview

This clause describes the structure for the Resource URIs and the resources and methods used for the service.

Figure 6.1.3.1-1 depicts the resource URIs structure for the Nadm_DM API.

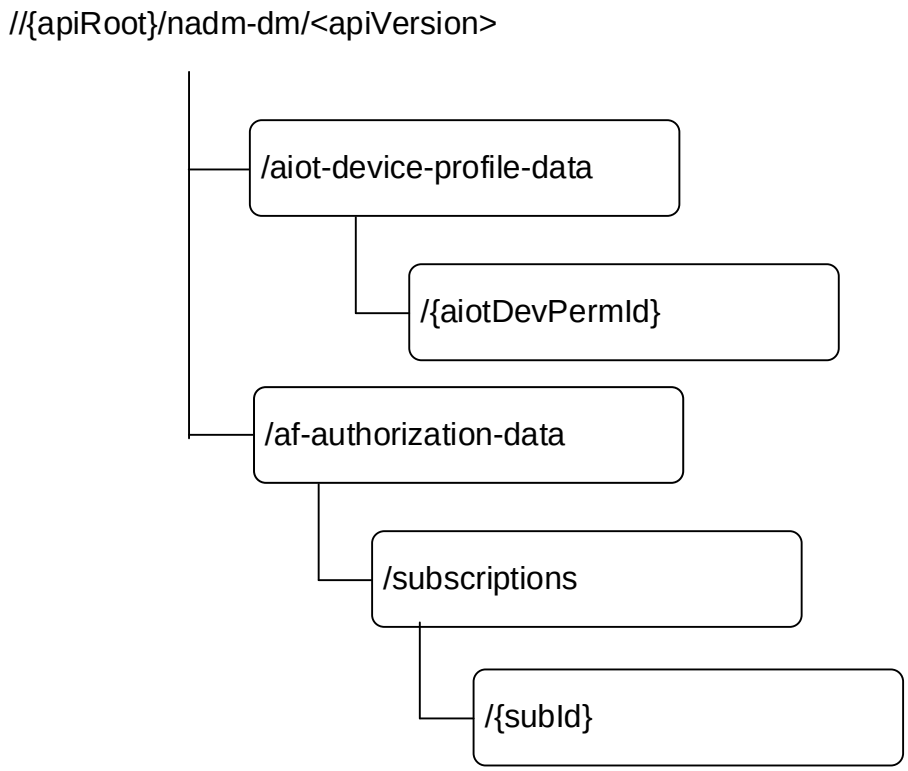


Figure 6.1.3.1-1: Resource URI structure of the Nadm_DM API

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.1.3.1-1: Resources and methods overview

Resource purpose/name	Resource URI (relative path after API URI)	HTTP method or custom operation	Description (service operation)
AiotDeviceProfileData	/aiot-device-profile-data/{aiotDevPermId}	GET	Retrieve AIoT Device Profile Data
		PATCH	Modify AIoT Device Profile Data
BundledAiotDeviceProfileData	/aiot-device-profile-data	GET	Retrieve AIoT Device Profile Data of bundled AIoT Devices.
		PATCH	Modify AIoT Device Profile Data of bundled AIoT Devices.
AfAuthorizationData	/af-authorization-data	GET	Retrieve AF Authorization Data
AfAuthorizationDataSubscriptions	/af-authorization-data/subscriptions	POST	Create a subscription of AF Authorization Data change
Individual AfAuthorizationDataSubscription	/af-authorization-data/subscriptions/{subId}	PATCH	Modify a subscription of AF Authorization Data change
		DELETE	Delete a subscription of AF Authorization Data change

6.1.3.2 Resource: AiotDeviceProfileData

6.1.3.2.1 Description

This resource is used to represent AIoT Device Profile Data.

6.1.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nadm-dm/<apiVersion>/aiot-device-profile-data/{aiotDevPermId}**

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

Table 6.1.3.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1
aiotDevPermId	AiotDevPermId	Represents the AIoT Device Permanent Identifier (see TS 23.369 [14] clause 5.7.2)

6.1.3.2.3 Resource Standard Methods

6.1.3.2.3.1 GET

This method shall support the URI query parameters specified in table 6.1.3.2.3.1-1.

Table 6.1.3.2.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability
n/a					

This method shall support the request data structures specified in table 6.1.3.2.3.1-2 and the response data structures and response codes specified in table 6.1.3.2.3.1-3.

Table 6.1.3.2.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.1.3.2.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AiotDevProfileData	M	1	200 OK	Upon success, a response body containing the AIoT Device Profile Data shall be returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.2.3.1-4: Headers supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
If-None-Match	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.2
If-Modified-Since	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.3

Table 6.1.3.2.3.1-5: Headers supported by the 200 response code on this resource

Name	Data type	P	Cardinality	Description
Cache-Control	string	O	0..1	Cache-Control containing max-age, as described in IETF RFC 9111 [16], clause 5.2
ETag	string	O	0..1	Entity Tag, containing a strong validator, as described in IETF RFC 9110 [15], clause 8.8.3
Last-Modified	string	O	0..1	Timestamp for last modification of the resource, as described in IETF RFC 9110 [15], clause 8.8.2

Table 6.1.3.2.3.1-6: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected

Table 6.1.3.2.3.1-7: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected.

6.1.3.2.3.2 PATCH

This method shall support the URI query parameters specified in table 6.1.3.2.3.2-1.

Table 6.1.3.2.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
supported-features	SupportedFeatures	O	0..1	see TS 29.500 [4] clause 6.6

This method shall support the request data structures specified in table 6.1.3.2.3.2-2 and the response data structures and response codes specified in table 6.1.3.2.3.2-3.

Table 6.1.3.2.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
AiotDevProfileData	M	1	Contains the updated AIoT Device Profile Data

Table 6.1.3.2.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned. (NOTE 3)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. NOTE 3: If all the modification instructions in the PATCH request have been implemented, the ADM shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the ADM shall respond with PatchResult.				

Table 6.1.3.2.3.2-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected

Table 6.1.3.2.3.2-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected.

6.1.3.3 Resource: AfAuthorizationData

6.1.3.3.1 Description

This resource is used to represent AF Authorization Data.

6.1.3.3.2 Resource Definition

Resource URI: **{apiRoot}/nadm-dm/<apiVersion>/af-authorization-data**

This resource shall support the resource URI variables defined in table 6.1.3.3.2-1.

Table 6.1.3.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

6.1.3.3.3 Resource Standard Methods

6.1.3.3.3.1 GET

This method shall support the URI query parameters specified in table 6.1.3.3.3.1-1.

Table 6.1.3.3.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability
af-id	AfId	C	0..1	This parameter shall be present if the request is to retrieve the AF authorization data for a specific AF. When present, this parameter indicates the AF identifier of the target AF. When absent, it indicates that the Authorization Data for all the AFs are to be retrieved.	

This method shall support the request data structures specified in table 6.1.3.3.3.1-2 and the response data structures and response codes specified in table 6.1.3.3.3.1-3.

Table 6.1.3.3.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.1.3.3.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AfAuthorizationData	M	1	200 OK	Upon success, a response body containing the AF Authorization Data shall be returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.3.3.1-4: Headers supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
If-None-Match	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.2
If-Modified-Since	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.3

Table 6.1.3.3.3.1-5: Headers supported by the 200 response code on this resource

Name	Data type	P	Cardinality	Description
Cache-Control	string	O	0..1	Cache-Control containing max-age, as described in IETF RFC 9111 [16], clause 5.2
ETag	string	O	0..1	Entity Tag, containing a strong validator, as described in IETF RFC 9110 [15], clause 8.8.3
Last-Modified	string	O	0..1	Timestamp for last modification of the resource, as described in IETF RFC 9110 [15], clause 8.8.2

Table 6.1.3.3.3.1-6: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected

Table 6.1.3.3.3.1-7: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected.

6.1.3.4 Resource: BundledAiotDeviceProfileData

6.1.3.4.1 Description

This resource is used to represent a collection resource of different AiotDeviceProfileData with the resource URI "aiotDevPermId".

6.1.3.4.2 Resource Definition

Resource URI: **{apiRoot}/nadm-dm/<apiVersion>/aiot-device-profile-data**

This resource shall support the resource URI variables defined in table 6.1.3.4.2-1.

Table 6.1.3.4.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

6.1.3.4.3 Resource Standard Methods

6.1.3.4.3.1 GET

This method shall support the URI query parameters specified in table 6.1.3.4.3.1-1.

Table 6.1.3.4.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability
requester-aiot-devices-id	array(AiotDevPermId)	M	1..N	This IE shall contain a list of the requested AIoT device permanent identifier.	

This method shall support the request data structures specified in table 6.1.3.4.3.1-2 and the response data structures and response codes specified in table 6.1.3.4.3.1-3.

Table 6.1.3.4.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.1.3.4.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AiotDevProfileDataList	M	1	200 OK	Upon success, a response body containing the AIoT Device Profile Data for bundled AIoT Devices shall be returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: The mandatory HTTP error status code for the GET method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.4.3.1-4: Headers supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
If-None-Match	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.2
If-Modified-Since	string	O	0..1	Validator for conditional requests, as described in IETF RFC 9110 [15], clause 13.1.3

Table 6.1.3.4.3.1-5: Headers supported by the 200 response code on this resource

Name	Data type	P	Cardinality	Description
Cache-Control	string	O	0..1	Cache-Control containing max-age, as described in IETF RFC 9111 [16], clause 5.2
ETag	string	O	0..1	Entity Tag, containing a strong validator, as described in IETF RFC 9110 [15], clause 8.8.3
Last-Modified	string	O	0..1	Timestamp for last modification of the resource, as described in IETF RFC 9110 [15], clause 8.8.2

Table 6.1.3.4.3.1-6: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected

Table 6.1.3.4.3.1-7: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected.

6.1.3.4.3.2 PATCH

This method shall support the URI query parameters specified in table 6.1.3.4.3.2-1.

Table 6.1.3.4.3.2-1: URI query parameters supported by the PATCH method on this resource

Name	Data type	P	Cardinality	Description
supported-features	SupportedFeatures	O	0..1	see TS 29.500 [4] clause 6.6

This method shall support the request data structures specified in table 6.1.3.4.3.2-2 and the response data structures and response codes specified in table 6.1.3.4.3.2-3.

Table 6.1.3.4.3.2-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	It contains the list of changes to be made to the AIoT Device Profile Data for bundled AIoT Devices, according to the JSON PATCH format specified in IETF RFC 6902 [13].

Table 6.1.3.4.3.2-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Upon success, an empty response body shall be returned. (NOTE 3)
PatchResult	M	1	200 OK	Upon success, the execution report is returned. (NOTE 2)
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4]. NOTE 3: If all the modification instructions in the PATCH request have been implemented, the ADM shall respond with 204 No Content response; if some of the modification instructions in the PATCH request have been discarded, and the NF service consumer has included in the supported-feature query parameter the "PatchReport" feature number, the ADM shall respond with PatchResult.				

Table 6.1.3.4.3.2-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected

Table 6.1.3.4.3.2-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the Callback URI of the target NF Service Consumer (e.g. ADM) to which the request is redirected. For the case, when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the request is redirected.

6.1.3.5 Resource: AfAuthorizationDataSubscriptions

6.1.3.5.1 Description

This resource is used to represent the collection of Subscriptions for AF Authorization Data Changes.

6.1.3.5.2 Resource Definition

Resource URI: **{apiRoot}/nadm-dm/<apiVersion>/af-authorization-data/subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.5.2-1.

Table 6.1.3.5.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1

6.1.3.5.3 Resource Standard Methods

6.1.3.5.3.1 POST

This method shall support the URI query parameters specified in table 6.1.3.5.3.1-1.

Table 6.1.3.5.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 6.1.3.5.3.1-2 and the response data structures and response codes specified in table 6.1.3.5.3.1-3.

Table 6.1.3.5.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
AfAuthorizationDataSubscription	M	1	Contain the information of the AF Authorization Data Subscription to be created.

Table 6.1.3.5.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
AfAuthorizationDataSubscription	M	1	201 Created	Upon success, a response body containing the AF Authorization Data Subscription shall be returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.5.3.1-4: Headers supported by the 201 response code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created subscription, according to the structure: {apiRoot}/nadm-dm/<apiVersion>/af-authorization-data/subscriptions/{subId}

Table 6.1.3.5.3.1-5: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.1.3.5.3.1-6: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.1.3.6 Resource: Individual AfAuthorizationDataSubscriptions

6.1.3.6.1 Description

This resource is used to represent an Individual Subscription of AF Authorization Data Changes.

6.1.3.6.2 Resource Definition

Resource URI: **{apiRoot}/nadm-dm/<apiVersion>/af-authorization-data/subscriptions/{subId}**

This resource shall support the resource URI variables defined in table 6.1.3.6.2-1.

Table 6.1.3.6.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 6.1.1
{subId}	string	Subscription Id

6.1.3.6.3 Resource Standard Methods

6.1.3.6.3.1 PATCH

This method shall support the URI query parameters specified in table 6.1.3.6.3.1-1.

Table 6.1.3.6.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 6.1.3.6.3.1-2 and the response data structures and response codes specified in table 6.1.3.6.3.1-3.

Table 6.1.3.6.3.1-2: Data structures supported by the PATCH Request Body on this resource

Data type	P	Cardinality	Description
array(PatchItem)	M	1..N	Contain the patch instructions to be applied to the subscription.

Table 6.1.3.6.3.1-3: Data structures supported by the PATCH Response Body on this resource

Data type	P	Cardinality	Response codes	Description
PatchResult	M	1	200 OK	Upon partial success, a response body containing the patch result shall be returned.
n/a			204 No Content	Upon success, 204 response code shall be returned.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: The mandatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.6.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.1.3.6.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.1.3.6.3.1 DELETE

This method shall support the URI query parameters specified in table 6.1.3.6.3.1-1.

Table 6.1.3.6.3.1-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description	Applicability

This method shall support the request data structures specified in table 6.1.3.6.3.1-2 and the response data structures and response codes specified in table 6.1.3.6.3.1-3.

Table 6.1.3.6.3.1-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.1.3.6.3.1-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a.			204 No Content	Upon success, 204 response code shall be returnd.
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of TS 29.500 [4] also apply. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.3.6.3.1-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.1.3.6.3.1-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.1.4 Custom Operations without associated resources

In this release of this specification, no custom operations without associated resources are defined for the Nadm_DM Service.

6.1.5 Notifications

6.1.5.1 General

This clause specifies the notifications provided by the Nadm_DM service.

The delivery of notifications shall be supported as specified in clause 6.2 of TS 29.500 [4] for Server-initiated communication.

Table 6.1.5.1-1: Notifications overview

Notification	Callback URI	HTTP method or custom operation	Description (service operation)
AF Authorization Data Change Notification	{notifyCallbackUri} (NF Service Consumer provided callback reference)	POST	Notify changes of AF Authorization Data

6.1.5.2 Notify changes of AF Authorization Data

6.1.5.2.1 Description

If the NF Service Consumer (e.g AIOTF) has provided the callback URI for getting notified about changes of AF Authorization Data, the ADM shall notify the NF Service Consumer when the AF Authorization Data is changed.

6.1.5.2.2 Notification Definition

The POST method shall be used for AF Authorization Data Change Notification, and the URI shall be the callback reference provided by the NF Service Consumer during the subscription to this notification.

Callback URI: **{notifyCallbackUri}**

Support of URI query parameters is specified in table 6.1.5.2.2-1.

Table 6.1.5.2.2-1: URI query parameters supported by the POST method

Name	Data type	P	Cardinality	Description
n/a				

Support of request data structures is specified in table 6.1.5.2.2-2, and support of response data structures and response codes is specified in table 6.1.5.2-3.

Table 6.1.5.2.2-2: Data structures supported by the POST Request Body

Data type	P	Cardinality	Description
AfAuthorizationDataChangeNotification	M	1	Representation of AF Authorization Data Change Notification

Table 6.1.5.2.2-3: Data structures supported by the POST Response Body

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful notification of the AF Authorization Data Change Notification
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of TS 29.500 [4]). NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.1.5.2.2-4: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the notification is redirected

Table 6.1.5.2.2-5: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target NF (service) instance ID towards which the notification is redirected

6.1.6 Data Model

6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Nadm_DM service based interface protocol.

Table 6.1.6.1-1: Nadm_DM specific Data Types

Data type	Clause defined	Description	Applicability
AiotDevProfileData	6.1.6.2.2	AIoT Device Profile Data	
LastKnownAiotfInfo	6.1.6.2.3	Last Known AIoTF Information	
IndividualAfAuthorizationData	6.1.6.2.4	Individual AF Authorization Data	
AllowedTargetAiotDevice	6.1.6.2.5	Allowed Target AIoT Device	
AfAuthorizationData	6.1.6.2.6	AF Authorization Data	
AiotDevProfileDataList	6.1.6.2.8	AIoT Device Profile Data of bundled AIoT Devices	
AfAuthorizationDataSubscription	6.1.6.2.9	AF Authorization Data Subscription	
AfAuthorizationDataChangeNotification	6.1.6.2.10	AF Authorization Data Change Notification	
LastKnownAiotfInfoInd	6.1.6.3.2	Last known AIOTF Information	
AfId	6.1.6.3.2	AF ID	
AllowedServiceOperation	6.1.6.3.3	Allowed Service Operation	

Table 6.1.6.1-2 specifies data types re-used by the Nadm_DM service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nadm_DM service based interface.

Table 6.1.6.1-2: Nadm_DM re-used Data Types

Data type	Reference	Comments	Applicability
SupportedFeatures	TS 29.571 [17]	see TS 29.500 [4] clause 6.6	SupportedFeatures
AiotDevPermId	TS 29.571 [17]	AIoT device permanent identifier	
AiotArea	TS 29.571 [17]	Contains the AIoT Area	
NfInstanceId	TS 29.571 [17]	Represent a NF instance.	
IpAddr	TS 29.571 [17]	Represent the IP address	
Fqdn	TS 29.571 [17]	Represent Fully Qualified Domain Name	
ProblemDetails	TS 29.571 [17]	Used in error responses to provide more detailed information about an error.	
Uri	TS 29.571 [17]	Unified Resource Identifier	
DateTime	TS 29.571 [17]	Date and Time	
PatchItem	TS 29.571 [17]	Patch Item	
PatchResult	TS 29.571 [17]	Patch Result	
PatchItem	TS 29.571 [17]	Patch item of JSON PATCH	
PatchResult	TS 29.571 [17]	Patch result of JSON PATCH	

6.1.6.2 Structured data types

6.1.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.1.6.2.2 Type: AiotDevProfileData

Table 6.1.6.2.2-1: Definition of type AiotDevProfileData

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	M	1	Globally unique AIoT device permanent identifier	
lastKnownAiotfInfo	LastKnownAiotfInfo	M	1	Indicate the last known AIOTF that serves the AIoT device, or unknown.	

6.1.6.2.3 Type: LastKnownAiotfInfo

Table 6.1.6.2.3-1: Definition of type LastKnownAiotfInfo

Attribute name	Data type	P	Cardinality	Description	Applicability
lastKnownAiotfInfoInd	boolean	M	1	Indicate the last known AIOTF that serves the AIoT device is known, or unknown. true: Known; false: Unknown	
lastKnownAiotfId	NfInstanceId	O	0..1	This IE may be present when lastKnownAiotfInfoInd is set to true. Indicates the last known AIOTF instance (NOTE)	
lastKnownAiotfAddress	IpAddr	O	0..1	This IE may be present when lastKnownAiotfInfoInd is set to true. Indicates the IP address for last known AIOTF instance (NOTE)	
lastKnownAiotfFqdn	Fqdn	O	0..1	This IE may be present when lastKnownAiotfInfoInd is set to true. Indicates the FQDN for last known AIOTF instance (NOTE)	
NOTE: At least, one of lastKnownAiotfId, lastKnownAiotfAddress and lastKnownAiotfFqdn shall be included when lastKnownAiotfInfoInd is set to true.					

6.1.6.2.4 Type: IndividualAfAuthorizationData

Table 6.1.6.2.4-1: Definition of type IndividualAfAuthorizationData

Attribute name	Data type	P	Cardinality	Description	Applicability
afId	AfId	M	1	Identifier used to identify the AF.	
allowedArea	AiotArea	O	0..1	If present, it indicates the allowed area for the indicated AF to perform the AIoT service operations. If this attribute is absent, it means that the complete service area is allowed for the AF.	
allowedServiceOperations	array(AllowedServiceOperation)	O	1..N	If present, it indicates the list of allowed service operation(s) for the AF. If this attribute is absent, it means that all the service operations are allowed for the AF.	
allowedTargetAiotDevices	array(AllowedTargetAiotDevice)	O	1..N	If present, it indicates the list of allowed AIoT Device(s) for the AF. If this attribute is absent, it means that all the Target Aiot Devices are allowed for the AF.	

6.1.6.2.5 Type: AllowedTargetAiotDevice

Table 6.1.6.2.5-1: Definition of type AllowedTargetAiotDevice

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	C	0..1	Indicates the AIoT Device Permanent Identifier. (NOTE)	
filteringInfo	AiotFilteringInformation	C	0..1	Indicates the Filtering Information. (NOTE)	
NOTE: Either aiotDevPermId, or filteringInfo shall be present.					

6.1.6.2.6 Type: AfAuthorizationData

Table 6.1.6.2.6-1: Definition of type AfAuthorizationData

Attribute name	Data type	P	Cardinality	Description	Applicability
afAuthData	map(IndividualAfAuthorizationData)	M	1..N	A map contains the AF Authorization Data in the ADM. The key of the map is the AF ID and the value of the map is the Authorization data of the corresponding AF indicated by the key.	

6.1.6.2.7 Void

6.1.6.2.8 Type: AiotDevProfileDataList

Table 6.1.6.2.8-1: Definition of type AiotDevProfileDataList

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevProfileDataList	array(AiotDevProfileData)	M	1..N	This IE indicates the AIoT Device Profile Data of a list of AIoT devices.	

6.1.6.2.9 Type: AfAuthorizationDataSubscription

Table 6.1.6.2.9-1: Definition of type AfAuthorizationDataSubscription

Attribute name	Data type	P	Cardinality	Description
notifyCallbackUri	Uri	M	1	Indicates the callback reference on the NF service consumer to receive the AF Authorization Data Change Notification.
notifyCorrelationId	string	M	1	Indicates the notification correlation Id for the subscription.
afId	AfId	O	0..1	Indicate the specific AF for which the AF Authorization Data changes to be monitored. If absent, changes of AF Authorization Data for all the AFs are to be monitored.
expiry	DataTime	C	0..1	This IE shall be present in the service response to indicate the expiry time of the subscription determined by the ADM. This IE may be present in the service request to indicate the expected expiry time proposed by the NF service consumer.

6.1.6.2.10 Type: AfAuthorizationDataChangeNotification

Table 6.1.6.2.10-1: Definition of type AfAuthorizationDataChangeNotification

Attribute name	Data type	P	Cardinality	Description
updateAfAuthData	map(IndividualAfAuthorizationData)	M	1..N	A map contains the updated AF Authorization Data in the ADM. The key of the map is the AF ID and the value of the map is the Authorization data of the corresponding AF indicated by the key.
notifyCorrelationId	string	M	1	Indicates the notification correlation Id of the subscription.

6.1.6.3 Simple data types and enumerations

6.1.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.1.6.3.2 Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.

Table 6.1.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
AfId	string	Identifier used to identify the AF.	

6.1.6.3.3 Enumeration: AllowedServiceOperation

The enumeration AllowedServiceOperation represents the different types of the allowed service operation for the AF.

Table 6.1.6.3.3-1: Enumeration AllowedServiceOperation

Enumeration value	Description	Applicability
"INVENTORY"	Indicates inventory service operation is allowed for the AF.	
"READ"	Indicates read service operation is allowed for the AF.	
"WRITE"	Indicates write service operation is allowed for the AF.	
"PERMANENT_DISABLE"	Indicates permanent disable service operation is allowed for the AF.	

6.1.6.3.4 Void

6.1.7 Error Handling

6.1.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

6.1.7.2 Protocol Errors

Protocol errors handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

6.1.7.3 Application Errors

The common application errors defined in Table 5.2.7.2-1 in TS 29.500 [4] may be used for the Nadm_DM. The application errors defined for the Nadm_DM service are listed in Table 6.1.7.3-1.

Table 6.1.7.3-1: Application errors

Application Error	HTTP status code	Description
DATA_NOT_FOUND	404 Not Found	The requested AIoT Device Profile Data or AF Authorization Data is not found/does not exist.

6.1.8 Feature negotiation

The optional features in table 6.1.8-1 are defined for the Nadm_DM API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

Table 6.1.8-1: Supported Features

Feature number	Feature Name	Description

6.1.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nadm_DM API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nadm_DM API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nadm_DM service.

The Nadm_DM API defines the following scopes for OAuth2 authorization.

Table 6.1.9-1: Oauth2 scopes defined in Nadm_DM API

Scope	Description
"nadm-dm"	Access to the Nadm_DM API
"nadm-dm:aiot-device-profile-data:read"	Access to read AIoT Device Profile Data
"nadm-dm:aiot-device-profile-data:modify"	Access to update AIoT Device Profile Data
"nadm-dm:af-authorization-data:read"	Access to read AF Authorization Data
"nadm-dm:af-authorization-data:subscriptions:create"	Access to create the subscriptions of AF Authorization Data Changes
"nadm-dm:af-authorization-data:subscriptions:modify"	Access to modify or delete the subscriptions of AF Authorization Data Changes

6.1.10 HTTP redirection

An HTTP request may be redirected to a different ADM service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different ADM producer instance will return the NF Instance ID of the new ADM producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an ADM redirects a service request to a different ADM using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new ADM towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

6.2 Nadm_Sec Service API

6.2.1 Introduction

The Nadm_Sec shall use the Nadm_Sec API.

The API URI of the Nadm_Sec API shall be:

{apiRoot}/<apiName>/<apiVersion>

The request URIs used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of TS 29.501 [5], i.e.:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} shall be set as described in TS 29.501 [5].
- The <apiName> shall be "nadm-sec".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

6.2.2 Usage of HTTP

6.2.2.1 General

HTTP/2, IETF RFC 9113 [11], shall be used as specified in clause 5 of TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of TS 29.500 [4].

The OpenAPI [6] specification of HTTP messages and content bodies for the Nadm_Sec API is contained in Annex A.

6.2.2.2 HTTP standard headers

6.2.2.2.1 General

See clause 5.2.2 of TS 29.500 [4] for the usage of HTTP standard headers.

6.2.2.2.2 Content type

JSON, IETF RFC 8259 [12], shall be used as content type of the HTTP bodies specified in the present specification as specified in clause 5.4 of TS 29.500 [4]. The use of the JSON format shall be signalled by the content type "application/json".

"Problem Details" JSON object shall be used to indicate additional details of the error in a HTTP response body and shall be signalled by the content type "application/problem+json", as defined in IETF RFC 9457 [13].

6.2.2.3 HTTP custom headers

The mandatory HTTP custom header fields specified in clause 5.2.3.2 of TS 29.500 [4] shall be supported, and the optional HTTP custom header fields specified in clause 5.2.3.3 of TS 29.500 [4] may be supported.

6.2.3 Resources

6.2.3.1 Overview

There are no resources defined for this API in this release of the specification.

6.2.4 Custom Operations without associated resources

6.2.4.1 Overview

This clause describes the structure for custom operations without associated resources.

Figure 6.2.4.1-1 depicts the custom operation URIs structure for the Nadm_Sec API.

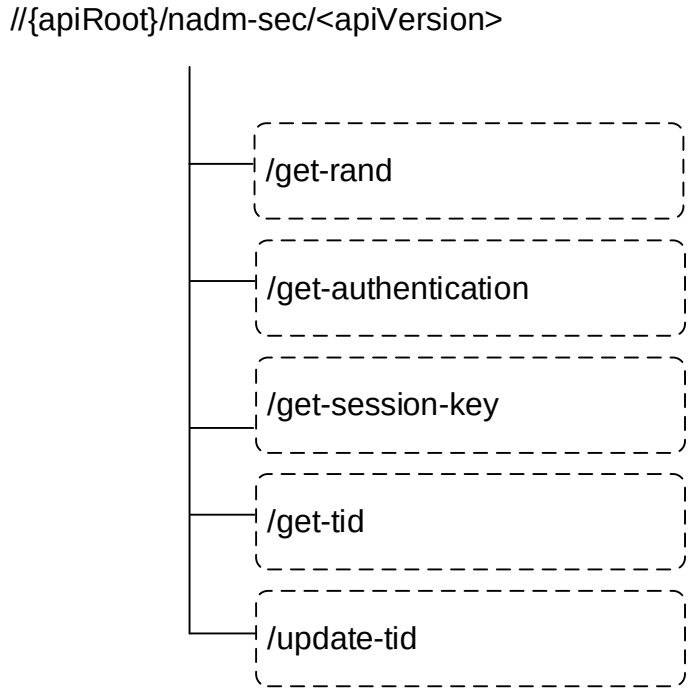


Figure 6.2.4.1-1: Custom operation URI structure of the Nadm_Sec API

Table 6.2.4.1-1 provides an overview of the custom operations and applicable HTTP methods.

Table 6.2.4.1-1: Custom operations without associated resources

Resource purpose/name	Resource URI (relative path after API URI)	HTTP method or custom operation	Description (service operation)
getRand	/get-rand	POST	Retrieve a random number from ADM
getAuthentication	/get-authentication	POST	Retrieve authentication data from ADM
getSessionKey	/get-session-key	POST	Retrieve a session key from ADM
getTid	/get-tid	POST	Retrieve a T-ID from ADM
updateTid	/update-tid	POST	Update a T-ID in ADM

6.2.4.2 Operation: getRand

6.2.4.2.1 Description

The custom operation enables to generate and retrieve a random number. See TS 33.369 [18].

6.2.4.2.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.4.2.2-1 and the response data structure and response codes specified in table 6.2.4.2.2-2.

Table 6.2.4.2.2-1: Data structures supported by the POST Request Body on this custom operation

Data type	P	Cardinality	Description
n/a			

Table 6.2.4.2.2-2: Data structures supported by the POST Response Body on this custom operation

Data type	P	Cardinality	Response codes	Description
GetRandResponse	M	1	200 OK	Upon success, a response body containing the generated random value $RAND_{AIOT_n}$ shall be returned
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.2.4.2.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.2.4.2.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.2.4.3 Operation: getAuthentication

6.2.4.3.1 Description

This custom operation enables to calculate and retrieve the authentication data. See TS 33.369 [18].

6.2.4.3.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.4.3.2-1 and the response data structure and response codes specified in table 6.2.4.3.2-2.

Table 6.2.4.3.2-1: Data structures supported by the POST Request Body on this custom operation

Data type	P	Cardinality	Description
GetAuthenticationRequest	M	1	Contains input parameters required to retrieve authentication data from ADM.

Table 6.2.4.3.2-2: Data structures supported by the POST Response Body on this custom operation

Data type	P	Cardinality	Response codes	Description
GetAuthenticationResponse	M	1	200 OK	Upon success, a response body containing the authentication data shall be returned
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
ProblemDetails	O	0..1	413 Payload Too Large	The "cause" attribute may be used to indicate one of the following application errors: - FILTER_INFORMATION_TOO_BROAD
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.2.4.3.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.2.4.3.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.2.4.4 Operation: getSessionKey

6.2.4.4.1 Description

This custom operation enables to calculate and retrieve the Session Key. See TS 33.369 [18].

6.2.4.4.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.4.4.2-1 and the response data structure and response codes specified in table 6.2.4.4.2-2.

Table 6.2.4.4.2-1: Data structures supported by the POST Request Body on this custom operation

Data type	P	Cardinality	Description
GetSessionKeyRequest	M	1	Contains input parameters required to retrieve the Session Key from ADM.

Table 6.2.4.4.2-2: Data structures supported by the POST Response Body on this custom operation

Data type	P	Cardinality	Response codes	Description
GetSessionKeyResponse	M	1	200 OK	Upon success, a response body containing the Session Key shall be returned
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.2.4.4.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.2.4.4.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.2.4.5 Operation: getTid

6.2.4.5.1 Description

This custom operation enables to derive a T-ID. See TS 33.369 [18].

6.2.4.5.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.4.5.2-1 and the response data structure and response codes specified in table 6.2.4.5.2-2.

Table 6.2.4.5.2-1: Data structures supported by the POST Request Body on this custom operation

Data type	P	Cardinality	Description
GetTidRequest	M	1	Contains input parameters for retrieving the T-ID.

Table 6.2.4.5.2-2: Data structures supported by the POST Response Body on this custom operation

Data type	P	Cardinality	Response codes	Description
GetTidResponse	M	1	200 OK	Upon success, a response body containing the T-ID information shall be returned
RedirectResponse	O	0..1	307 Temporary Redirect	Temporary redirection. (NOTE 2)
RedirectResponse	O	0..1	308 Permanent Redirect	Permanent redirection. (NOTE 2)
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE 1: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported. NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of TS 29.500 [4].				

Table 6.2.4.5.2-3: Headers supported by the 307 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

Table 6.2.4.5.2-4: Headers supported by the 308 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	An alternative URI of the resource located on an alternative service instance within the same ADM or ADM (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in TS 29.500 [4].
3gpp-Sbi-Target-Nf-Id	string	O	0..1	Identifier of the target ADM (service) instance ID towards which the request is redirected

6.2.4.6 Operation: updateTid

6.2.4.6.1 Description

This custom operation enables to update a T-ID in ADM. See TS 33.369 [18].

6.2.4.6.2 Operation Definition

This operation shall support the request data structures specified in table 6.2.4.6.2-1 and the response data structure and response codes specified in table 6.2.4.6.2-2.

Table 6.2.4.6.2-1: Data structures supported by the POST Request Body on this custom operation

Data type	P	Cardinality	Description
UpdateTidRequest	M	1	Contains input parameters required to update a T-ID in ADM.

Table 6.2.4.6.2-2: Data structures supported by the POST Response Body on this custom operation

Data type	P	Cardinality	Response codes	Description
UpdateTidResponse	M	1	200 OK	Upon success, a response body containing the T-ID information shall be returned.
ProblemDetails	O	0..1	404 Not Found	The "cause" attribute may be used to indicate one of the following application errors: - DATA_NOT_FOUND
NOTE: In addition common data structures as listed in table 5.2.7.1-1 of TS 29.500 [4] are supported.				

6.2.5 Notifications

6.2.6 Data Model

6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Nadm_Sec service based interface protocol.

Table 6.2.6.1-1: Nadm_Sec specific Data Types

Data type	Clause defined	Description	Applicability
GetAuthenticationRequest	6.2.6.2.2	Contains input parameters for computing authentication data	
AuthData	6.2.6.2.3	Authentication Data	
AuthDataSet	6.2.6.2.4	Authentication Data Set	
GetAuthenticationResponse	6.2.6.2.5	Contains authentication data	
GetSessionKeyRequest	6.2.6.2.6	Contains input parameters for computing a session key	
GetSessionKeyResponse	6.2.6.2.7	Contains a session key information	
GetTidRequest	6.2.6.2.8	Contains input parameters for retrieving the T-ID	
GetTidResponse	6.2.6.2.9	Contains T-ID information	
GetRandResponse	6.2.6.2.10	Contains RAND information	
TidHandlingInformation	6.2.6.2.11	T-ID Handling Information	
UpdateTidRequest	6.2.6.2.12	Contains input parameters required to update a T-ID in ADM.	
UpdateTidResponse	6.2.6.2.13	Contains T-ID information.	
Kaiotf	6.2.6.3.2	AIoT session key K_{AIOTF}	
Tid	6.2.6.3.2	AIoT Device Temporary Identifier (T-ID)	
TidType	6.2.6.3.3	T-ID Type	

Table 6.2.6.1-2 specifies data types re-used by the Nadm_Sec service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nadm_Sec service based interface.

Table 6.2.6.1-2: Nadm_Sec re-used Data Types

Data type	Reference	Comments	Applicability
SupportedFeatures	TS 29.571 [17]	see TS 29.500 [4] clause 6.6	SupportedFeatures
AiotDevPermId	TS 29.571 [17]	AIoT device permanent identifier	
AiotFilteringInformation	TS 29.571 [17]	AIoT Filtering Information	
ProblemDetails	TS 29.571 [17]	Used in error responses to provide more detailed information about an error.	
Rand	TS 29.503 [19]	RAND	
Xres	TS 29.503 [19]	XRES	

6.2.6.2 Structured data types

6.2.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.2.6.2.2 Type: GetAuthenticationRequest

Table 6.2.6.2.2-1: Definition of type GetAuthenticationRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	O	0..1	Indicates the AIoT device permanent identifier.	
filteringInfo	AiotFilteringInformation	O	0..1	Indicates the AIoT Filtering Information.	
randN	Rand	M	1	Indicates the $RAND_{AIOT_n}$	
randDlist	array(Rand)	M	1..N	Indicates the $RAND_{AIOT_d}$(s)	

6.2.6.2.3 Type: AuthData

Table 6.2.6.2.3-1: Definition of type AuthData

Attribute name	Data type	P	Cardinality	Description	Applicability
xres	Xres	M	1	Contains the calculated XRES.	
aiotDevPermId	AiotDevPermId	C	0..1	This attribute shall be present if the aiotDevPermId attribute is not provided in the GetAuthenticationRequest data type. Indicates the AIoT device permanent identifier used to calculate the XRES.	

6.2.6.2.4 Type: AuthDataSet

Table 6.2.6.2.4-1: Definition of type AuthDataSet

Attribute name	Data type	P	Cardinality	Description	Applicability
randD	Rand	M	1	Indicates the $RAND_{AIOT_d}$ used to derive the authentication data	
authDataSet	array(AuthData)	M	1..N	Contains the list of calculated XRES.	

6.2.6.2.5 Type: GetAuthenticationResponse

Table 6.2.6.2.5-1: Definition of type GetAuthenticationResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
authDataSets	array(AuthDataSet)	M	1..N	Indicates the list of Authentication Data Set	

6.2.6.2.6 Type: GetSessionKeyRequest

Table 6.2.6.2.6-1: Definition of type GetSessionKeyRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	M	1	Indicates an AIoT device permanent identifier. (NOTE)	
randN	Rand	M	1	Indicates the $RAND_{AIOT_n.}$	
randD	Rand	M	1	Indicates the $RAND_{AIOT_d.}$	

6.2.6.2.7 Type: GetSessionKeyResponse

Table 6.2.6.2.7-1: Definition of type GetSessionKeyResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
kAiotf	Kaiotf	M	1	Contains the Session Key	

6.2.6.2.8 Type: GetTidRequest

Table 6.2.6.2.8-1: Definition of type GetTidRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	M	1	Indicates an AIoT device permanent identifier.	
resyncInd	boolean	O	0..1	Indicates if re-synchronization is required. true: re-synchronization is required; false (or absent): re-synchronization is not required;	
randN	Rand	O	0..1	Indicates the RANDAIOT_n	

6.2.6.2.9 Type: GetTidResponse

Table 6.2.6.2.9-1: Definition of type GetTidResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
tidHandInfo	TidHandlingInformation	M	1	Indicate T-ID Handling Information	
tid	Tid	M	1	Current Temporary Identifier (T-ID_n) as specified in TS 33.369 [18]	

6.2.6.2.10 Type: GetRandResponse

Table 6.2.6.2.10-1: Definition of type GetRandResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
randN	Rand	M	1	Indicates the $RAND_{AIOT_n}$	

6.2.6.2.11 Type: TidHandlingInformation

Table 6.1.6.2.11-1: Definition of type TidHandlingInformation

Attribute name	Data type	P	Cardinality	Description	Applicability
tidType	TidType	M	1	Indicates the T-ID handling type.	
tidTypeUpdateInd	boolean	C	0..1	This IE shall be present if the TidType set to "STORED". Indicates whether the stored T-ID is updated with a command via a Command procedure or without a command. true: indicates that the stored T-ID is updated with a command via a Command procedure. false: indicates that the stored T-ID is updated without a command.	

6.2.6.2.12 Type: UpdateTidRequest

Table 6.2.6.2.12-1: Definition of type UpdateTidRequest

Attribute name	Data type	P	Cardinality	Description	Applicability
aiotDevPermId	AiotDevPermId	M	1	Indicates an AIoT device permanent identifier.	
randN	Rand	M	1	Indicates a $RAND_{AIOT_n}$.	

6.2.6.2.13 Type: UpdateTidResponse

Table 6.2.6.2.13-1: Definition of type UpdateTidResponse

Attribute name	Data type	P	Cardinality	Description	Applicability
tid	Tid	M	1	Updated Temporary Identifier (T-ID) as specified in TS 33.369 [18]	

6.2.6.3 Simple data types and enumerations

6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

Table 6.2.6.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
Kaiotf	string	pattern: "^[A-Fa-f0-9]{32}$"	
Tid	string	pattern: "^[A-Fa-f0-9]{32}$"	

6.2.6.3.3 Enumeration: TidType

Table 6.2.6.3.3-1: Enumeration TidType

Enumeration value	Description
"STORED"	Indicates the stored type of T-ID
"CONCEALED"	Indicates the concealed type of T-ID

6.2.7 Error Handling

6.2.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of TS 29.500 [4].

6.2.7.2 Protocol Errors

Protocol errors handling shall be supported as specified in clause 5.2.7 of TS 29.500 [4].

6.2.7.3 Application Errors

The common application errors defined in Table 5.2.7.2-1 in TS 29.500 [4] may be used for the Nadm_Sec. The application errors defined for the Nadm_Sec service are listed in Table 6.2.7.3-1.

Table 6.2.7.3-1: Application errors

Application Error	HTTP status code	Description
DATA_NOT_FOUND	404 Not Found	The AIoT device permanent identifier or Filtering Information is not recognized.
FILTER_INFORMATION_TOO_BROAD	413 Payload Too Large	The group of devices derived from the Filtering Information is too large to compute the expected authentication data in a reasonable amount of time.

6.2.8 Feature negotiation

The optional features in table 6.2.8-1 are defined for the Nadm_Sec API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of TS 29.500 [4].

Table 6.2.8-1: Supported Features

Feature number	Feature Name	Description

6.2.9 Security

As indicated in TS 33.501 [8] and TS 29.500 [4], the access to the Nadm_Sec API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [9]), based on local configuration, using the "Client Credentials" authorization grant, where the NRF (see TS 29.510 [10]) plays the role of the authorization server.

If OAuth2 is used, an NF Service Consumer, prior to consuming services offered by the Nadm_Sec API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in TS 29.510 [10], clause 5.4.2.2.

NOTE: When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nadm_Sec service.

The Nadm_Sec API defines the following scopes for OAuth2 authorization.

Table 6.2.9-1: OAuth2 scopes defined in Nadm_Sec API

Scope	Description
"nadm-sec"	Access to the Nadm_Sec API
"nadm-sec:get-rand:invoke "	Access to retrieve a RAND
"nadm-sec:get-authentication:invoke "	Access to retrieve Authentication Data
"nadm-sec:get-session-key:invoke "	Access to retrieve a Session Key
"nadm-sec:get-tid:invoke "	Access to retrieve a T-ID
"nadm-sec:update-tid:invoke"	Access to update a T-ID in ADM

6.2.10 HTTP redirection

An HTTP request may be redirected to a different ADM service instance when using direct or indirect communications (see TS 29.500 [4]).

An SCP that reselects a different ADM producer instance will return the NF Instance ID of the new ADM producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of TS 29.500 [4].

If an ADM redirects a service request to a different ADM using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new ADM towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of TS 29.500 [4].

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the API(s) defined in the present specification. It consists of OpenAPI specifications in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

NOTE 1: The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository that uses the GitLab software version control system (see clause 5.3.1 of TS 29.501 [5] and clause 5B of TR 21.900 [7]).

A.2 Nadm_DM API

```
openapi: 3.0.0

info:
  version: '1.0.2'
  title: 'Nadm_DM'
  description: |
    Nadm Data Management Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.369 Ambient IoT Data Management Services, version 19.3.0
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.369/'

servers:
  - url: '{apiRoot}/nadm-dm/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501.

security:
  - oAuth2ClientCredentials:
    - nadm-dm
  - {}

paths:
  /aiot-device-profile-data/{aiotDevPermId}:
    get:
      summary: get AIoT Device Profile Data for a give AIoT Device
      operationId: Get AIoT Device Profile Data
      tags:
        - AIoT Device Profile Data Retrieval
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-dm
        - oAuth2ClientCredentials:
          - nadm-dm
          - nadm-dm:aiot-device-profile-data:read
      parameters:
        - name: aiotDevPermId
          in: path
          description: AIoT Device Permanent Identifier
          required: true
          schema:
```

```
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AiotDevProfileData'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        description: Not Found
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  patch:
    summary: modify AIoT Device Profile Data for a give AIoT Device
    operationId: Modify AIoT Device Profile Data
    tags:
      - AIoT Device Profile Data Update
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-dm
      - oAuth2ClientCredentials:
        - nadm-dm
        - nadm-dm:aiot-device-profile-data:modify
    parameters:
      - name: aiotDevPermId
        in: path
        description: AIoT Device Permanent Identifier
        required: true
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
      - name: supported-features
        in: query
        description: Features required to be supported by the target NF
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    requestBody:
      content:
        application/json-patch+json:
          schema:
            $ref: '#/components/schemas/AiotDevProfileData'
      required: true
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '204':
        description: Expected response to a valid request
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
```

```
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        description: Not Found
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error

  /aiot-device-profile-data:
    get:
      summary: get AIoT Device Profile Data for a bundled AIoT Devices
      operationId: Get Bundled AIoT Device Profile Data
      tags:
        - Bundled AIoT Device Profile Data Retrieval
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-dm
        - oAuth2ClientCredentials:
          - nadm-dm
          - nadm-dm:aiot-device-profile-data:read
      parameters:
        - name: requester-aiot-devices-id
          in: query
          description: a list of AIoT Device Permanent Identifier
          required: true
          style: form
          explode: false
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
            minItems: 1
      responses:
        '200':
          description: Expected response to a valid request
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/AiotDevProfileDataList'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '406':
          $ref: 'TS29571_CommonData.yaml#/components/responses/406'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
```

```
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error

  patch:
    summary: modify bundled AIoT Device Profile Data for bundled AIoT devices.
    operationId: Modify bundled AIoT Device Profile Data
    tags:
      - Bundled AIoT Device Profile Data Update
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-dm
      - oAuth2ClientCredentials:
        - nadm-dm
        - nadm-dm:aiot-device-profile-data:modify
    parameters:
      - name: supported-features
        in: query
        description: Features required to be supported by the target NF
        schema:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    requestBody:
      content:
        application/merge-patch+json:
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
      required: true
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '204':
        description: Expected response to a valid request
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error

/af-authorization-data:
  get:
    summary: get AF Authorization Data
    operationId: Get AF Authorization Data
    tags:
```

```
      - AF Authorization Data Retrieval
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-dm
      - oAuth2ClientCredentials:
        - nadm-dm
        - nadm-dm:af-authorization-data:read
    parameters:
      - name: af-id
        in: query
        description: AF ID
        required: false
        schema:
          $ref: '#/components/schemas/AfId'
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AfAuthorizationData'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        description: Not Found
        content:
          application/problem+json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
      '406':
        $ref: 'TS29571_CommonData.yaml#/components/responses/406'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error

  /af-authorization-data/subscriptions:
    post:
      summary: Create a subscription of AF Authorization Data Changes
      operationId: Subscribe AF Authorization Data Change
      tags:
        - Create a subscription of AF Authorization Data Changes
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-dm
        - oAuth2ClientCredentials:
          - nadm-dm
          - nadm-dm:af-authorization-data:subscriptions:create
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AfAuthorizationDataSubscription'
        required: true
      callbacks:
        afAuthorizationDataChangeNotification:
          '{$request.body#/notifyCalblackUri}':
            post:
              requestBody:  # contents of the callback message
                required: true
                content:
                  application/json:
```

```
                  schema:
                    $ref: '#/components/schemas/AfAuthorizationDataChangeNotification'
            responses:
              '204':
                description: successful notification
              '307':
                $ref: 'TS29571_CommonData.yaml#/components/responses/307'
              '308':
                $ref: 'TS29571_CommonData.yaml#/components/responses/308'
              '400':
                $ref: 'TS29571_CommonData.yaml#/components/responses/400'
              '403':
                $ref: 'TS29571_CommonData.yaml#/components/responses/403'
              '404':
                $ref: 'TS29571_CommonData.yaml#/components/responses/404'
              '411':
                $ref: 'TS29571_CommonData.yaml#/components/responses/411'
              '413':
                $ref: 'TS29571_CommonData.yaml#/components/responses/413'
              '415':
                $ref: 'TS29571_CommonData.yaml#/components/responses/415'
              '429':
                $ref: 'TS29571_CommonData.yaml#/components/responses/429'
              '500':
                $ref: 'TS29571_CommonData.yaml#/components/responses/500'
              '502':
                $ref: 'TS29571_CommonData.yaml#/components/responses/502'
              '503':
                $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    responses:
      '201':
        description: successful creation of the subscription
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/AfAuthorizationDataSubscription'
        headers:
          Location:
            description: >
              'Contains the URI of the newly created subscription, according to the structure:
              {apiRoot}/nadm-dm/<apiVersion>/af-authorization-data/subscriptions/{subId}'
            required: true
            schema:
              type: string
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  /af-authorization-data/subscriptions/{subId}:
    patch:
      summary: Modify a subscription of AF Authorization Data Changes
```

```
    operationId: Modify Subscription of AF Authorization Data Change
    tags:
      - Modify a subscription of AF Authorization Data Changes
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-dm
      - oAuth2ClientCredentials:
        - nadm-dm
        - nadm-dm:af-authorization-data:subscriptions:modify
    requestBody:
      content:
        application/json-patch+json:
          schema:
            type: array
            items:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
      required: true
    responses:
      '200':
        description: Expected response to partially successful patch
        content:
          application/json:
            schema:
              $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchResult'
      '204':
        description: Expected response to successful patch
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

  delete:
    summary: Delete a subscription of AF Authorization Data Changes
    operationId: Unsubscribe to AF Authorization Data Change
    tags:
      - Delete a subscription of AF Authorization Data Changes
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-dm
      - oAuth2ClientCredentials:
        - nadm-dm
        - nadm-dm:af-authorization-data:subscriptions:modify
    responses:
      '204':
        description: Expected response to successful deletion
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
```

```
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '501':
          $ref: 'TS29571_CommonData.yaml#/components/responses/501'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nadm-dm: Access to the Nadm_DM API
            nadm-dm:aiot-device-profile-data:read: Access to read AIoT Device Profile Data
            nadm-dm:aiot-device-profile-data:modify: Access to update AIoT Device Profile Data
            nadm-dm:af-authorization-data:read: Access to read AF Authorization Data
            nadm-dm:af-authorization-data:subscriptions:create: Access to create the subscriptions
of AF Authorization Data Changes
            nadm-dm:af-authorization-data:subscriptions:modify: Access to modify or delete the
subscriptions of AF Authorization Data Changes

  schemas:
#
# STRUCTURED TYPES
#

    AiotDevProfileData:
      description: Contains the AIoT Device Profile Data.
      type: object
      required:
        - aiotDevPermId
        - lastKnownAiotfInfo
      properties:
        aiotDevPermId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
        lastKnownAiotfInfo:
          $ref: '#/components/schemas/LastKnownAiotfInfo'

    LastKnownAiotfInfo:
      description: Contains the Last Known AIoTF Info.
      type: object
      required:
        - lastKnownAiotfInfoInd
      properties:
        lastKnownAiotfInfoInd:
          type: boolean
        lastKnownAiotfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
        lastKnownAiotfAddress:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/IpAddr'
        lastKnownAiotfFqdn:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Fqdn'

    IndividualAfAuthorizationData:
      description: Contains the AF Authorization Data for an individual AF.
      type: object
      required:
        - afId
      properties:
        afId:
          $ref: '#/components/schemas/AfId'
        allowedArea:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotArea'
        allowedServiceOperations:
```

```
          type: array
          items:
            $ref: '#/components/schemas/AllowedServiceOperation'
          minItems: 1
        allowedTargetAiotDevices:
          type: array
          items:
            $ref: '#/components/schemas/AllowedTargetAiotDevice'
          minItems: 1

    AllowedTargetAiotDevice:
      description: >
        Contains the permanent AIoT Device ID or the filtering information,
        either aiotDevPermId, or filteringInfo shall be present.
      type: object
      properties:
        aiotDevPermId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
        filteringInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotFilteringInformation'

    AfAuthorizationData:
      description: Contains the AF Authorization Data.
      type: object
      required:
        - afAuthData
      properties:
        afAuthData:
          type: object
          description: >
            'Map carrying the AF Authorization Data.
            The key of the map is the AF ID of the corresponding Authorization Data'
          additionalProperties:
            $ref: '#/components/schemas/IndividualAfAuthorizationData'
          minProperties: 1

    AfAuthorizationDataSubscription:
      description: Contains a AF Authorization Data Change Subscription.
      type: object
      required:
        - notifyCallbackUri
        - notifyCorrelationId
      properties:
        notifyCallbackUri:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
        notifyCorrelationId:
          type: string
        afId:
          $ref: '#/components/schemas/AfId'
        expiry:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'

    AfAuthorizationDataChangeNotification:
      description: Contains a AF Authorization Data Change Notification.
      type: object
      required:
        - udpateAfAuthData
        - notifyCorrelationId
      properties:
        udpateAfAuthData:
          type: object
          description: >
            'Map carrying the Updated AF Authorization Data.
            The key of the map is the AF ID of the corresponding Authorization Data'
          additionalProperties:
            $ref: '#/components/schemas/IndividualAfAuthorizationData'
          minProperties: 1
        notifyCorrelationId:
          type: string

#
# SIMPLE TYPES
#

    AfId:
      description: Indicates the identifier used to identify the AF.
      type: string
```

```
    AiotDevProfileDataList:
      description: AIoT Device Profile Data of a list of AIoT devices.
      type: object
      required:
        - aiotDevProfileDataList
      properties:
        aiotDevProfileDataList:
          type: array
          items:
            $ref: '#/components/schemas/AiotDevProfileData'
          minItems: 1


#
# ENUMS
#

    AllowedServiceOperation:
      description: The Allowed Service Operation.
      anyOf:
        - type: string
          enum:
            - INVENTORY
            - READ
            - WRITE
            - PERMANENT_DISABLE
        - type: string
```

A.3 Nadm_Sec API

```
openapi: 3.0.0

info:
  version: '1.0.2'
  title: 'Nadm_Sec'
  description: |
    Nadm_Sec Service.
    © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.

externalDocs:
  description: 3GPP TS 29.369 Ambient IoT Data Management Services, version 19.3.0
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.369/'

servers:
  - url: '{apiRoot}/nadm-sec/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501.

security:
  - oauth2ClientCredentials:
    - nadm-sec
  - {}

paths:
  /get-rand:
    post:
      summary: Retrieve a random value (RANDAIOT_n)
      operationId: getRand
      tags:
        - Get Rand
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-sec
        - oAuth2ClientCredentials:
          - nadm-sec
          - nadm-sec:get-rand:invoke
      responses:
        '200':
          description: Successful retrieval of RANDAIOT_n
```

```
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/GetRandResponse'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
/get-authentication:
  post:
    summary: Retrieve authentication data (XRESAIOT)
    operationId: getAuthentication
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/GetAuthenticationRequest'
    tags:
      - Get Authentication
    security:
      - {}
      - oAuth2ClientCredentials:
        - nadm-sec
      - oAuth2ClientCredentials:
        - nadm-sec
        - nadm-sec:get-authentication:invoke
    responses:
      '200':
        description: Successful retrieval of authentication data
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/GetAuthenticationResponse'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
```

```
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  /get-session-key:
    post:
      summary: Retrieve session key (KAIOTF)
      operationId: getSessionKey
      tags:
        - Get Session Key
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-sec
        - oAuth2ClientCredentials:
          - nadm-sec
          - nadm-sec:get-session-key:invoke
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/GetSessionKeyRequest'
      responses:
        '200':
          description: Successful retrieval of session key
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/GetSessionKeyResponse'
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
  /get-tid:
    post:
      summary: Retrieve Temporary Identifier (T-ID)
      operationId: getTid
      tags:
        - Get Tid
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-sec
        - oAuth2ClientCredentials:
          - nadm-sec
          - nadm-sec:get-tid:invoke
      requestBody:
        required: true
        content:
```

```
        application/json:
          schema:
            $ref: '#/components/schemas/GetTidRequest'
    responses:
      '200':
        description: Successful retrieval of T-ID
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/GetTidResponse'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      default:
        description: Unexpected error
  /update-tid:
    post:
      summary: Update Temporary ID (T-ID) in ADM
      operationId: updateTid
      tags:
        - Update Tid
      security:
        - {}
        - oAuth2ClientCredentials:
          - nadm-sec
        - oAuth2ClientCredentials:
          - nadm-sec
          - nadm-sec:update-tid:invoke
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UpdateTidRequest'
      responses:
        '200':
          description: Successful update of T-ID
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/UpdateTidResponse'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          description: Not Found
          content:
            application/problem+json:
              schema:
                $ref: 'TS29571_CommonData.yaml#/components/schemas/ProblemDetails'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
```

```
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          description: Unexpected error
components:
  securitySchemes:
    oauth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: https://example.com/oauth2/token
          scopes:
            nadm-sec: Access to Nadm_Sec API
            nadm-sec:get-rand:invoke: Access to retrieve a RAND
            nadm-sec:get-authentication:invoke: Access to retrieve Authentication Data
            nadm-sec:get-session-key:invoke: Access to retrieve a Session Key
            nadm-sec:get-tid:invoke: Access to retrieve a T-ID
            nadm-sec:update-tid:invoke: Access to update a T-ID in ADM
  schemas:
#
# STRUCTURED TYPES
#

    GetRandResponse:
      type: object
      description: Contains the RandN.
      properties:
        randN:
          $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
      required:
      - randN

    GetAuthenticationRequest:
      type: object
      description: Contains the Rand and AIoT identifications.
      properties:
        aiotDevPermId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
        filteringInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotFilteringInformation'
        randN:
          $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
        randDlist:
          type: array
          items:
            $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
          minItems: 1
      required:
      - randN
      - randDlist

    AuthData:
      type: object
      description: >
        Contains the calculated XRES and optionally the AIoT device permanent
        identifier used to calculate it.
      properties:
        xres:
          $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Xres'
        aiotDevPermId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
      required:
      - xres

    AuthDataSet:
      type: object
      description: Contains a RANDAIOT_d value and a list of associated AuthData entries.
      properties:
```

```
      randD:
        $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
      authDataSet:
        type: array
        items:
          $ref: '#/components/schemas/AuthData'
        minItems: 1
    required:
    - randD
    - authDataSet

  GetAuthenticationResponse:
    type: object
    description: Response type containing one or more AuthDataSet entries.
    properties:
      authDataSets:
        type: array
        items:
          $ref: '#/components/schemas/AuthDataSet'
        minItems: 1
    required:
    - authDataSets

  GetSessionKeyRequest:
    type: object
    description: Contains a AIoT Device Permanent Identifier, RandN and RandD.
    properties:
      aiotDevPermId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
      randN:
        $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
      randD:
        $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
    required:
    - aiotDevPermId
    - randN
    - randD

  GetSessionKeyResponse:
    type: object
    description: Contains the Kaiotf.
    properties:
      kAiotf:
        $ref: '#/components/schemas/Kaiotf'
    required:
    - kAiotf

  GetTidRequest:
    type: object
    description: Contains a AIoT Device Permanent Identifier and Resynchronization Indicator.
    properties:
      aiotDevPermId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
      resyncInd:
        type: boolean
      randN:
        $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
    required:
    - aiotDevPermId

  GetTidResponse:
    type: object
    description: Contains T-ID Handling Info and T-ID.
    properties:
      tidHandInfo:
        $ref: '#/components/schemas/TidHandlingInformation'
      tid:
        $ref: '#/components/schemas/Tid'
    required:
    - tid
    - tidHandInfo

  TidHandlingInformation:
    description: Contains the T-ID handling information
    type: object
    required:
      - tidType
    properties:
```

```
        tidType:
          $ref: '#/components/schemas/TidType'
        tidTypeUpdateInd:
          type: boolean

    Tid:
      description: AIoT Device Temporary Identifier (T-ID)
      type: string
      pattern: '^[A-Fa-f0-9]{32}$'

    UpdateTidRequest:
      type: object
      properties:
        aiotDevPermId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AiotDevPermId'
        randN:
          $ref: 'TS29503_Nudm_UEAU.yaml#/components/schemas/Rand'
      required:
      - aiotDevPermId
      - randN

    UpdateTidResponse:
      type: object
      description: Contains updated T-ID.
      properties:
        tid:
          $ref: '#/components/schemas/Tid'
      required:
      - tid

#
# SIMPLE TYPES
#

    Kaiotf:
      description: Indicates the Kaiotf.
      type: string
      pattern: '^[A-Fa-f0-9]{32}$'


#
# ENUMS
#
    TidType:
      description: Indicates the type of T-ID.
      anyOf:
        - type: string
          enum:
            - STORED
            - CONCEALED
        - type: string
```

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2025-04	CT4#128	C4-251315 C4-251316 C4-251317 C4-251318 C4-251460				TS skeleton Implementation of pCRs agreed at CT4#128.	0.1.0
2025-05	CT4#129	C4-252212 C4-252270 C4-252335 C4-252339 C4-252436 C4-252337 C4-252365				TS skeleton Implementation of pCRs agreed at CT4#129.	0.2.0
2025-08	CT4#130	C4-253305 C4-253406 C4-253407 C4-253408 C4-253409 C4-253410 C4-253411				TS skeleton Implementation of pCRs agreed at CT4#130.	0.3.0
2025-09	CT#109	CP-252182				Presented for information and approval	1.0.0
2025-09	CT#109					Approved in TSG CT#109	19.0.0
2025-12	CT#110	CP-253160	0001		D	Editorial Correction	19.1.0
2025-12	CT#110	CP-253160	0002	1	F	Remove the incorrect NF consumer	19.1.0
2025-12	CT#110	CP-253160	0003		F	Correction of OpenAPI	19.1.0
2025-12	CT#110	CP-253160	0004	1	F	Missing HTTP Response Code	19.1.0
2025-12	CT#110	CP-253160	0005	7	B	AIoT support for T-IDs	19.1.0
2025-12	CT#110	CP-253190	0008	3	B	New ADM security service	19.1.0
2025-12	CT#110	CP-253226	0014	3	B	Support T-ID handling information	19.1.0
2025-12	CT#110	CP-253154	0015	2	F	Correction of attribute presence	19.1.0
2025-12	CT#110	CP-253167	0016		F	29.369 Rel-19 API version and External doc update	19.1.0
2026-03	CT#111	CP-260036	0017	1	F	Correction on T-ID	19.2.0
2026-03	CT#111	CP-260036	0018		F	Add the descriptions	19.2.0
2026-03	CT#111	CP-260036	0019		D	Editorial Corrections	19.2.0
2026-03	CT#111	CP-260036	0020	1	F	Update on OpenAPI for get-rand	19.2.0
2026-03	CT#111	CP-260036	0021		F	Update on RandGetResponse	19.2.0
2026-03	CT#111	CP-260036	0022	1	F	Update on security parts	19.2.0
2026-03	CT#111	CP-260036	0023	1	F	Update on the length of Kaiotf	19.2.0
2026-03	CT#111	CP-260036	0025	2	F	Add bundled AIoT Device Profile Data retrieval and update procedures	19.2.0
2026-03	CT#111	CP-260036	0027	1	F	Make permanent ID a conditional parameter in Authentication_Get response	19.2.0
2026-03	CT#111	CP-260036	0029		F	Correction for supporting paging all	19.2.0
2026-03	CT#111	CP-260036	0031		F	Nadm_SecTID_Get correction	19.2.0
2026-03	CT#111	C4-260400	0032		F	29.369 Rel-19 API version and External doc update	19.2.0
2026-06	CT#112	C4-262215	0028	5	F	New T-ID Update Service Operation	19.3.0
2026-06	CT#112	C4-261362	0033	1	F	Correction to Nadm_Sec	19.3.0
2026-06	CT#112	C4-261364	0035	1	F	Procedure Description for Batch Retrieval of AF Authorization Data	19.3.0
2026-06	CT#112	C4-261142	0038		F	Correction on Schema for AIoT Device Profile Data Bulk Modification	19.3.0
2026-06	CT#112	C4-261143	0039		F	Missing Response Codes in Nadm_Sec service operations	19.3.0
2026-06	CT#112	C4-262146	0040	2	F	Nadm DM Subscription Operations	19.3.0
2026-06	CT#112	C4-261177	0045		F	Terms alignment	19.3.0
2026-06	CT#112	C4-262105	0054		F	Add missing reference	19.3.0
2026-06	CT#112	C4-262143	0055		F	Separate T-ID from Device Profile Data	19.3.0
2026-06	CT#112	C4-262453	0056		F	29.369 Rel-19 API version and External doc update	19.3.0

History

Version	Date	Status
V19.0.0	January 2026	Publication
V19.1.0	February 2026	Publication
V19.2.0	March 2026	Publication
V19.3.0	August 2026	Publication